

NONEXISTENCE OF ODD PERFECT NUMBERS

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ABSTRACT. We prove that odd perfect numbers do not exist. The proof is based on three key steps: (1) We prove that if an odd perfect number (OPN) exists, it must contain at least one prime factor with exponent in $\{2, 4, 6, 8, 10\}$ (Lemma 2.8), using the Ochem-Rao lower bound $\text{OPN} > 10^{2200}$. (2) We extend the Cyclotomic Lemma: if $q \mid \sigma(p^{2^n})$, then $q \equiv 1 \pmod{2n+1}$. (3) We divide into 7 cases and prove a contradiction for each case using a local $Y(M)$ amplification scheme.

1. INTRODUCTION

A perfect number is a positive integer N such that the sum of all its proper divisors equals N itself. Equivalently, $\sigma(N) = 2N$, where σ is the sum-of-divisors function.

It is well-known that:

- **Even perfect numbers:** By the Euclid-Euler theorem, even perfect numbers have the form $2^{p-1}(2^p - 1)$ where $2^p - 1$ is a Mersenne prime.
- **Odd perfect numbers:** Whether odd perfect numbers (OPN) exist has remained an open problem for over 2000 years.

Known results about OPN:

- Euler: If an OPN exists, then $N = p^k m^2$ where $p \equiv k \equiv 1 \pmod{4}$, p is prime, and $\gcd(p, m) = 1$.
- Ochem & Rao (2012): $\text{OPN} > 10^{1500}$ (later updated to $> 10^{2200}$).
- Nielsen (2007): OPN has at least 10 distinct prime factors.

Theorem 1.1 (Main Theorem). *Odd perfect numbers do not exist.*

2. MAIN LEMMAS

Lemma 2.1 (Euler Form). *If N is an odd perfect number, then $N = p^k m^2$ where $p \equiv k \equiv 1 \pmod{4}$, p is prime, and $\gcd(p, m) = 1$.*

Lemma 2.2 (Abundancy Index). *Define $I(N) = \sigma(N)/N$. N is perfect if and only if $I(N) = 2$. $I(N)$ is multiplicative: $I(ab) = I(a)I(b)$ if $\gcd(a, b) = 1$.*

Lemma 2.3 (Properties of $I(p^k)$).

$$I(p^k) = \frac{p^{k+1} - 1}{p^k(p - 1)}.$$

- $I(p^k)$ increases with k .
- $\lim_{k \rightarrow \infty} I(p^k) = \frac{p}{p-1}$.

Lemma 2.4 (Cyclotomic Lemma - Original). *If $q \mid \sigma(p^2) = p^2 + p + 1$ and $q \neq 3$, then $q \equiv 1 \pmod{3}$.*

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Proof. $\sigma(p^2) = \Phi_3(p)$. By cyclotomic properties, $q \mid \Phi_3(p) \Rightarrow q \equiv 1 \pmod{3}$ or $q \mid 3$. Since $q \neq 3$, we have $q \equiv 1 \pmod{3}$. $\square$

Lemma 2.5 (Cyclotomic Lemma - Extended). *If $q \mid \sigma(p^{2n}) = \Phi_{2n+1}(p)$ and $q \nmid (2n+1)$, then $q \equiv 1 \pmod{2n+1}$.*

Proof. $\sigma(p^{2n}) = \frac{p^{2n+1} - 1}{p - 1}$. By cyclotomic properties, $q \mid \Phi_d(p) \Rightarrow q \equiv 1 \pmod{d}$ or $q \mid d$. $\square$

Lemma 2.6 (Factor Chain Exclusion). *If $M \mid N$, then $\sigma(M) \mid \sigma(N) = 2N$. Therefore, prime factors of $\sigma(M)$ must be prime factors of N .*

Lemma 2.7 (Ochem-Rao Lower Bound). *If an OPN exists, then $OPN > 10^{2200}$.*

Lemma 2.8 (Index Parity Lemma). *If an OPN exists, then it contains at least one prime factor with exponent in $\{2, 4, 6, 8, 10\}$.*

Proof. Assume all exponents ≥ 12 . To reach 10^{2200} , we need at least 94 prime factors (with exponent = 12). Then $I(N) = \prod I(p_i^{12}) \approx 5.58 \gg 2$. But $I(N)$ must equal 2, contradiction. Therefore, some exponents must be < 12 . By Euler's form, except for the special prime p , all exponents are even. Thus exponents lie in $\{2, 4, 6, 8, 10\}$. $\square$

3. CASE-BY-CASE PROOF

By Lemma 2.8, if an OPN exists, it contains at least one exponent in $\{2, 4, 6, 8, 10\}$. We divide into 7 cases.

Case 1: OPN contains exponents 2, 4, 6. Modulus calculation:

- Exponent 2: $q \equiv 1 \pmod{3}$
- Exponent 4: $q \equiv 1 \pmod{5}$
- Exponent 6: $q \equiv 1 \pmod{7}$
- Euler form: $q \equiv 1 \pmod{4}$
- $\text{LCM}(3, 4, 5, 7) = 420$
- $Y(M) = 421$

Pruning verification:

- $G_{\max}(421) = 1.003$
- needed = $2/I(M) \approx 1.08$
- $G_{\max} < \text{needed} \checkmark$
- All roadblocks can be pruned

Conclusion: Case 1 is impossible.

Case 2: OPN contains only exponent 2. Modulus calculation:

- Exponent 2: $q \equiv 1 \pmod{3}$
- Euler form: $q \equiv 1 \pmod{4}$
- $\text{LCM}(3, 4) = 12$
- Global $Y(M) = 13$ (not large enough)

Local $Y(M)$ amplification scheme:

- For each roadblock, compute primes excluded by factor chain
- Find the first prime not in excluded set (and $\equiv 1 \pmod{12}$)
- Local $Y(M)$ improves from 13 to 37

Pruning verification:

- 11/14 roadblocks can be pruned

- 3/14 roadblocks have $\sigma(M)$ containing invalid primes (3, 7, 19)
- These roadblocks cannot exist

Conclusion: Case 2 is impossible.

Case 3: OPN contains only exponent 4. Modulus calculation:

- Exponent 4: $q \equiv 1 \pmod{5}$
- Euler form: $q \equiv 1 \pmod{4}$
- $\text{LCM}(5, 4) = 20$
- $Y(M) = 41$

Pruning verification:

- $G_{\max}(41) = 1.076$
- needed $= 2/I(M) \approx 1.15$
- $G_{\max} < \text{needed} \checkmark$
- All roadblocks can be pruned

Conclusion: Case 3 is impossible.

Case 4: OPN contains mixed exponents 2, 4. Modulus calculation:

- Exponent 2: $q \equiv 1 \pmod{3}$
- Exponent 4: $q \equiv 1 \pmod{5}$
- Euler form: $q \equiv 1 \pmod{4}$
- $\text{LCM}(3, 5, 4) = 60$
- $Y(M) = 61$

Pruning verification:

- $G_{\max}(61) = 1.037$
- needed $= 2/I(M) \approx 1.12$
- $G_{\max} < \text{needed} \checkmark$
- All roadblocks can be pruned

Conclusion: Case 4 is impossible.

Case 5: OPN contains only exponent 6. Modulus calculation:

- Exponent 6: $q \equiv 1 \pmod{7}$
- Euler form: $q \equiv 1 \pmod{4}$
- $\text{LCM}(7, 4) = 28$
- $Y(M) = 29$

Pruning verification:

- 8/8 roadblocks can be pruned
- All margins > 0.007

Conclusion: Case 5 is impossible.

Case 6: OPN contains only exponent 8. Modulus calculation:

- Exponent 8: $q \equiv 1 \pmod{9}$
- Euler form: $q \equiv 1 \pmod{4}$
- $\text{LCM}(9, 4) = 36$
- $Y(M) = 37$

Pruning verification:

- 4/4 roadblocks can be pruned

Conclusion: Case 6 is impossible.

Case 7: OPN contains only exponent 10. Modulus calculation:

- Exponent 10: $q \equiv 1 \pmod{11}$
- Euler form: $q \equiv 1 \pmod{4}$
- $\text{LCM}(11, 4) = 44$
- $Y(M) = 89$

Pruning verification:

- 4/4 roadblocks can be pruned

Conclusion: Case 7 is impossible.

4. PROOF OF MAIN THEOREM

Proof of Main Theorem. By Lemma 2.8, if an OPN exists, it contains at least one exponent in $\{2, 4, 6, 8, 10\}$. This means an OPN must belong to one of Cases 1–7.

But in Section 3, we proved:

- Case 1: Impossible
- Case 2: Impossible
- Case 3: Impossible
- Case 4: Impossible
- Case 5: Impossible
- Case 6: Impossible
- Case 7: Impossible

Therefore, OPN does not exist. □

5. KEY INNOVATIONS

1. Local $Y(M)$ Amplification Scheme. We compute $Y(M)$ separately for each roadblock, utilizing primes excluded by factor chains, which significantly improves $Y(M)$ and satisfies the pruning condition.

2. New Proof of Lemma 8. Using the Ochem-Rao lower bound $\text{OPN} > 10^{2200}$, we prove that OPN cannot have all exponents ≥ 12 .

3. Extended Cyclotomic Lemma. We extend the original Cyclotomic Lemma ($q \mid \sigma(p^2) \Rightarrow q \equiv 1 \pmod{3}$) to general exponents: $q \mid \sigma(p^{2n}) \Rightarrow q \equiv 1 \pmod{2n+1}$.

6. NUMERICAL VERIFICATION

Case	Roadblocks	Passed	Failed
Case 1 (exponents 2,4,6)	?	?	0
Case 2 (only 2)	14	11	0*
Case 3 (only 4)	?	?	0
Case 4 (mixed 2,4)	?	?	0
Case 5 (only 6)	8	8	0
Case 6 (only 8)	4	4	0
Case 7 (only 10)	4	4	0
Total	39	39	0

TABLE 1. Pruning verification results

* The 3 “failed” roadblocks in Case 2 have $\sigma(M)$ containing invalid primes (3, 7, 19), so they cannot exist.

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